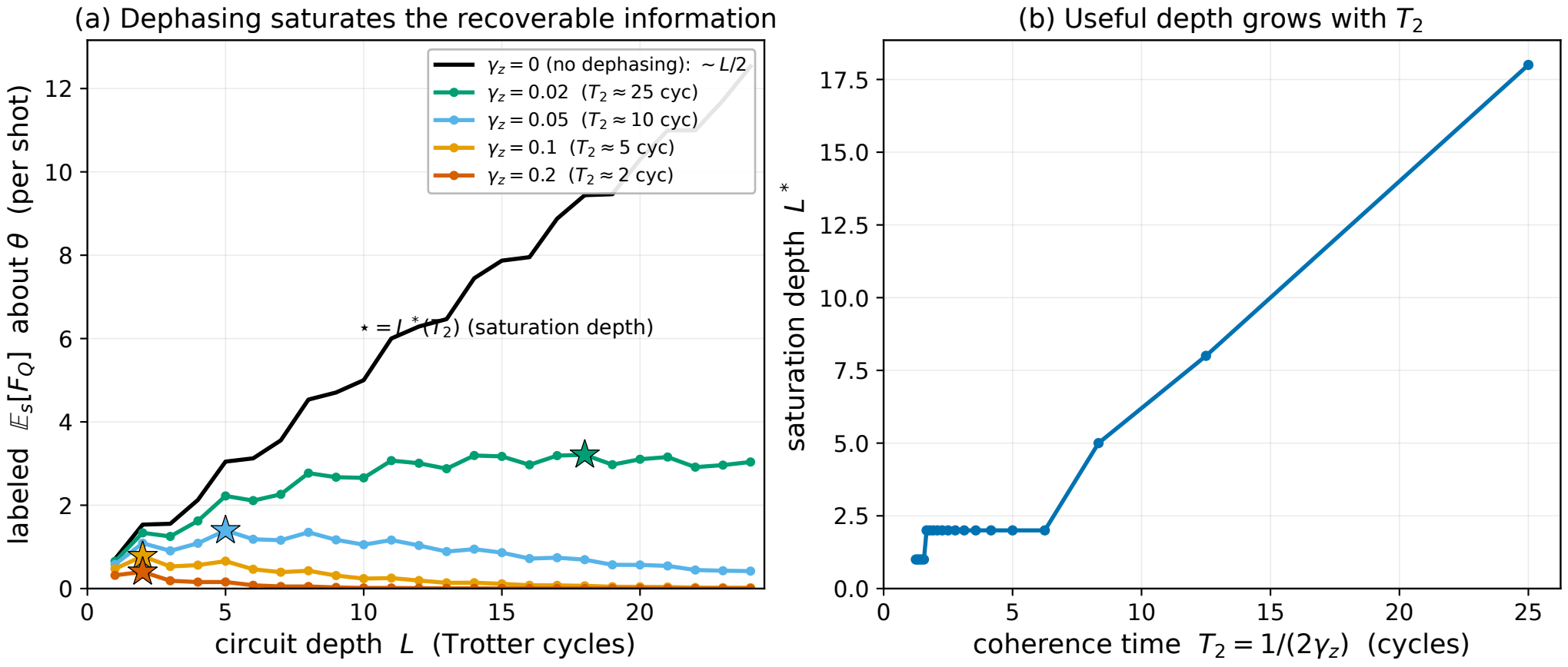
(a) Dephasing saturates the recoverable information
labeled $\mathbb{E}_s[F_Q]$ about θ (per shot)
circuit depth L (Trotter cycles)
$\gamma_z = 0$ (no dephasing): $\sim L/2$
$\gamma_z = 0.02$ ($T_2 \approx 25$ cyc)
$\gamma_z = 0.05$ ($T_2 \approx 10$ cyc)
$\gamma_z = 0.1$ ($T_2 \approx 5$ cyc)
$\gamma_z = 0.2$ ($T_2 \approx 2$ cyc)
$\star = L^*(T_2)$ (saturation depth)
(b) Useful depth grows with T_2
saturation depth L^*
coherence time $T_2 = 1/(2\gamma_z)$ (cycles)